

g) Apply page replacement algo. for following seq of pages. FIFO & check whether beladi's anomaly exists or not. For checking anomaly take phy. mem. size as 3 frames & 4 frames.

1 2 3 4 1 2 5 1 2 3 4 5.

FIFO:

		3	3	3	2	2	2	2	2	2	5
	2	2	2	1	1		1	1	1	1	4
1	1	1	4	4	4	5	5	5	3	3	3
x	x	x	x	x	x	x	✓	✓	x	x	✓

Page fault = 9

$$\text{hit} = \frac{3}{12} = 0.25 = 25\%$$

1	2	3	4	1	2	5	1	2	3	4	5
			4	4	4	4	4	2	2	4	4
	3	3	3	3	3	3	3	3	3	3	3
	2	2	2	2	2	1	1	1	1	2	
1	1	1	1	1	1	5	5	5	5	1	1
x	x	x	x	✓	✓	x	x	x	x	x	x

faults (hits) = 10

hits = 2

$$\text{hit ratio} = 16.67\% \left(\frac{2}{12} \times 100 \right)$$

Belady's anomaly : Since increasing the physical memory from 3 frames to 4 frames, inc page faults from 9 to 10, Belady's anomaly exists.

8) For the following sequence of pairs find hit ratio using
 FIFO LRU MRU optimal page replacement algo.
 with physical mem. of 3 frames.

FIFO:

3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
1	1	1	1	1	2	2	2	2	2	1	1	1	1	1	1	1	1	1
2	2	2	2	6	6	6	6	4	4	4	4	4	2	2	2	2	2	2
3	3	3	3	4	4	4	4	4	3	3	3	3	3	5	5	5	5	4
X	X	X	✓	X	✓	X	X	✓	X	X	✓	X	X	X	X	X	X	X

Hit ratio = $\frac{6}{19} \times 100 = 31.58\%$. Miss = $1 - \frac{6}{19} = \frac{13}{19}$.

2) LRU:

1	1	1	1	1	1	1	1	4	4	4	4	4	4	4	4	4	1	1
2	2	2	4	4	4	2	2	2	2	2	2	2	2	5	5	5	3	3
3	3	3	3	3	3	6	6	6	3	3	3	3	3	1	1	1	2	2
X	X	X	✓	X	✓	X	X	X	X	✓	✓	X	✓	X	X	X	X	X

Hits = $\frac{5}{19} \times 100 = 26.32\%$.

Miss = $1 - \frac{5}{19} = \frac{14}{19}$.

MRU:

3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
		1	1	1	1	6	6	6	6	6	6	6	6	6	6	6	6	6
	2	2	2	2	2	2	2	2	2	2	2	1	1	1	1	1	3	4
3	3	3	3	4	4	4	4	4	3	4	4	4	4	5	2	2	2	2
X	X	X	✓	X	✓	X	✓	✓	X	X	✓	X	✓	X	X	✓	F	F

$$\text{Hits} = 7$$

$$\text{Hit} = \frac{7}{19}$$

$$\text{Miss} = 1 - \frac{7}{19} = \frac{12}{19}$$

④

Optimal

			1	1	1	1	6	6	6	3	3	3	1	1	1	1	1	1	
	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	
3	3	3	3	4	4	4	4	4	4	4	4	4	4	4	5	5	5	3	4
X	X	X	✓	X	✓	X	✓	✓	X	✓	✓	X	✓	X	✓	✓	✓	X	X

$$\text{Hit} = \frac{9}{19}$$

$$\text{Miss} = 1 - \frac{9}{19} = \frac{10}{19}$$

For this question, based on hit ratio, best option

is optimal page replacement.

because it replaces the page that will not be used for the longest time in the future.